INTERNATIONAL MONETARY FUND

MALDIVES

April 30, 2019

STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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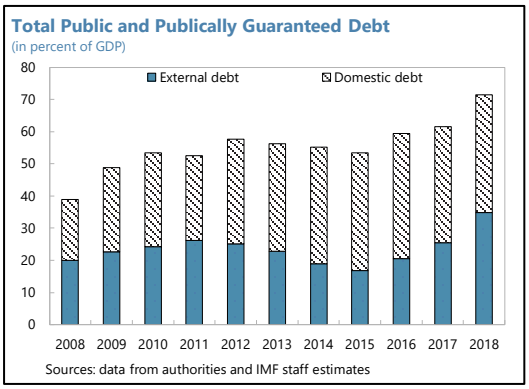
Prepared by the staffs of the International Monetary Fund and the International Development Association.

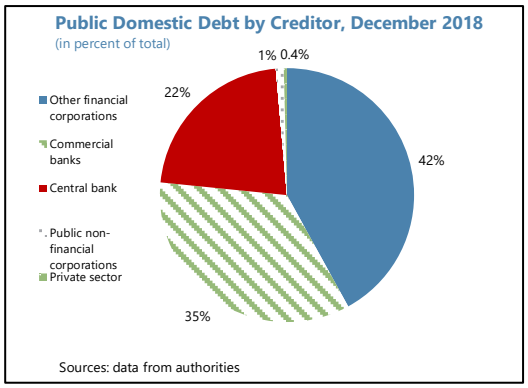
Maldives Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress:	High
Overall risk of debt distress:	High
Granularity in the risk rating:	Sustainable
Application of judgement:	No

The risk of external debt distress in the Maldives remains high. External public and publicly-guaranteed debt increased by 10 percent of GDP in 2018 leading all indicators, except the debt-to-exports ratio, to breach their thresholds under the baseline. The authorities are predominantly financing their infrastructure investment scale-up with external debt, some of which is guaranteed. The debt profile is susceptible to rollover risks from outstanding international sovereign bonds and financing pressures from amortizations of guaranteed loans for major infrastructure projects. A Sovereign Development Fund has been set up with resources allocated to mitigate anticipated liquidity pressures. This along with the winding down of large-scale infrastructure supports an assessment that debt is sustainable. Public domestic debt remains elevated and around half of it is composed of short-term Treasury bills. Fiscal policy slippages and shocks to tourism exports and capital expenditures remain key risks. Stabilizing and reducing public debt ratios will require strengthening the medium-term fiscal framework to improve the external position.

BACKGROUND AND DEVELOPMENTS ON DEBT

1. Total public and publicly-guaranteed (PPG) debt in Maldives increased by 10 percent of GDP in 2018 and stood at around US$3.8 billion in 2018, approximately 72 percent of GDP. The increase in public sector debt was due to external PPG debt associated with major infrastructure projects. Disbursements in 2018 were exceptionally high because of an overlap of ending (i.e., bridge, airport runway, housing) and starting (notably, new housing development and airport terminal) projects. About 50 percent of public sector debt is domestic and most of it is denominated in local currency. Domestic debt is held by financial institutions including finance and insurance companies, pension funds, and other fund accounts (42 percent), commercial banks (35 percent), and the Maldives Monetary Authority (MMA) (22 percent).[1] Short-term domestic debt accounts for around half of total domestic debt. The domestic debt market is not fully developed, and the authorities have received TA on formulating a Medium-Term Debt Management Strategy (MTDS).[2] They plan to issue securities with longer maturities, though work on this has not begun.





2. External PPG debt recorded around US$1.9 billion in 2018. The recent expansion in infrastructure investment has been financed with external debt. Investment in transportation, housing, and electricity has aimed at filling an infrastructure gap and increasing growth potential, especially from tourism. Around 60 percent of external PPG debt is owed to bilateral official creditors.[3] Of bilateral debt, 83 percent is owed to China. This is equivalent to US$928 million or about 50 percent of PPG external debt, of which US$403 million is guaranteed. Guaranteed external debt increased markedly in 2018 with the bulk owed to China in association with housing projects carried out by the Housing Development Corporation, a state-owned enterprise (SOE). The terms on most guaranteed debt are non-concessional.[4] Sovereign bonds were issued in 2017 and 2018, totaling US$350 million (6.6 percent of 2018 GDP), on commercial

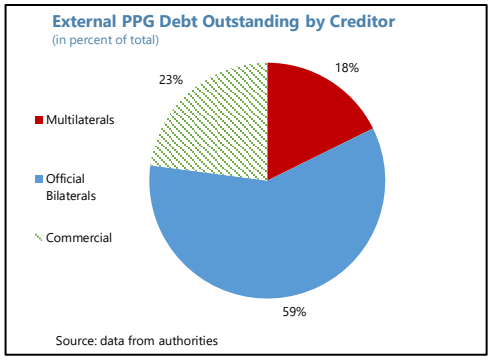
[1] This is a conversion of the outstanding debt held by the MMA in the Ways and Means account into a long-term Treasury bond in December 2014.

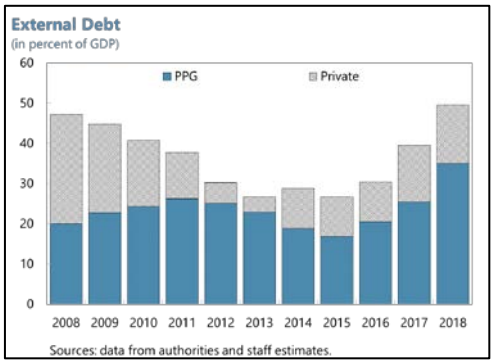
[2] A Debt Management Performance Assessment (DeMPA) evaluation is being finalized and the results of the assessment are expected to help the Government of Maldives (GoM) to design a medium-term reform plan.

[3] Debt outstanding to EXIM China and EXIM India (buyers' credit lending) is included in bilateral debt. External PPG debt to bilateral creditors is owed mostly to China, 83 percent, followed by other non-Paris Club members, 8 percent.

[4] Debt outstanding from state-owned Chinese banks is categorized as bilateral, at non-concessional terms, given that the grant element is below 35 percent.

lending terms. External debt is mostly denominated in U.S. dollar (53 percent), renminbi (19 percent), and SDR (17 percent).





3. The Fund's debt sustainability analysis includes PPG external [5] and domestic debt. Public debt includes debt of the central government, including guarantees to SOEs, and debt of the central bank. Public debt does not include non-guaranteed debt of SOEs as they can borrow without the guarantee of the government.

Subsectors of the public sector	Sub-sectors covered
Central government	X
State and local government	
Other elements in the general government	
o/w: Social security fund	
o/w: Extra budgetary funds (EBFs)	
Guarantees (to other entities in the public and private sector, including to SOEs)	X
Central bank (borrowed on behalf of the government)	X
Non-guaranteed SOE debt	

The country's coverage of public debt	The central government plus social security, central bank, government-guaranteed debt			
	Default		Used for the analysis	Reasons for deviations from the default settings
Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0	
PPP	35	percent of PPP stock	0.0	
Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0	
Total (2+3+4+5) (in percent of GDP)			7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND ON MACRO FORECASTS

4. Despite strong growth and stable inflation, external imbalances in the Maldives have worsened compared to the last Debt Sustainability Analysis (DSA).

[5] External debt in this DSA is defined by currency criterion.

- **Current account (CAD).** The non-interest CAD is projected at 23 percent of GDP in 2018, around 8 percent of GDP higher than previously projected. The larger deficit is due to higher imports related to increased implementation of large infrastructure projects, which was largely financed by an increase in PPG external debt. Because of the higher CAD, the adjustment over the medium term is more accelerated as SOE projects come to an end. CAD adjustment also benefits from higher tourist arrivals which are projected to grow at 8 percent over the medium term compared to 5.5 percent.

- **Fiscal balance.** Preliminary outturns for 2018 shows a lower primary deficit reflecting mainly lower current spending compared to the previous DSA – this includes lower spending on salaries and allowances, administrative services, and social welfare. However, given the absence of new revenue measures and the assumption of a more moderate decline in capital spending, the current fiscal projection shows a slower medium-term decline in the primary deficit compared to the previous DSA. Domestic financing is projected to cover around 60 percent of gross financing needs over the medium term. Maldives will continue to receive grants from bilateral and multilateral partners, while the magnitude is smaller than concessional borrowing and declines over the medium-term.

Change in Macro Assumptions

	2018	2019	2020	2021	2022	2023
Real GDP growth (in percent change)						
Current	7.5	6.5	6.0	5.5	5.5	5.5
Previous	4.7	4.8	5.0	5.0	5.0	4.6
Inflation						
Current	2.0	2.0	2.4	2.5	2.5	2.3
Previous	2.0	2.0	2.4	2.5	2.5	2.3
Primary fiscal deficit (in percent of GDP)						
Current	2.6	3.0	3.2	2.6	2.2	1.8
Previous	4.6	3.3	2.5	1.9	0.8	0.9
Non-interest current account deficit (in percent of GDP)						
Current	23.0	18.0	13.0	11.2	9.3	8.0
Previous	15.4	14.7	12.5	11.0	8.4	9.0

- **Growth and inflation.** Growth was higher than anticipated in 2018 due to strong performance in the tourism sector and a pickup in construction related to large infrastructure projects. Over the medium term, growth is slightly higher than projected in the last DSA driven by improvements in the tourism sector against better infrastructure.

5. The new realism tools suggest that the macroeconomic projections are reasonable and consistent with historical patterns and those observed in other LICs. The three-year change in the primary balance is at the median of the distribution of planned fiscal adjustments in the sample of Fund-supported programs in low-income countries since 1990. The primary deficit will increase relative to the trough in 2017. Planned public investment is greater than previously projected as implementation rates for existing projects have increased. The larger contribution of government capital to projected growth, compared to the previous DSA, reflects greater public investment. Current projections are reasonably close

to estimates of the historical contribution. Growth in 2019 is projected lower than values implied by a range of fiscal multipliers because construction on infrastructure projects, financed outside the budget and not reflected in the fiscal deficit, will slow down.

EXTERNAL AND PUBLIC DEBT SUSTAINABILITY

6. The debt carrying capacity of the Maldives remains weak. Maldives' Composite Indicator (CI) score is calculated to 2.48, and the country has 'weak debt carrying capacity'. CI is based on a weighted average of several factors such as the country's real GDP growth, remittances, international reserves, and world growth and the CPIA score, and the calculation of the CI is based on 10-year averages of the variables, across 5 years of historical data and 5 years of projection, and the corresponding CPIA.

Debt Carrying Capacity	Weak

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Weak	Weak 2.48	Weak 2.44	Weak 3.18

Note: Until the April 2019 WEO vintage is released, the two previous vintages ago classification and corresponding score are based solely on the CPIA per the previous framework.

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.177	1.22	49%
Real growth rate (in percent)	2.719	5.231	0.14	6%
Import coverage of reserves (in percent)	4.052	19.344	0.78	32%
Import coverage of reserves^2 (in percent)	-3.990	3.742	-0.15	-6%
Remittances (in percent)	2.022	0.000	0.00	0%
World economic growth (in percent)	13.520	3.579	0.48	19%
CI Score			2.48	100%
CI rating			Weak	

7. External debt sustainability

- **PPG external debt-to-GDP** stood at around 35 percent in 2018 and is projected to increase to 41 percent of GDP by 2021 before declining to 33 percent in 2028. Strong FDI flows put external debt on a declining path over the medium term despite a CAD. FDI flows to the Maldives have been historically strong, averaging 9 percent of GDP between 2007 and 2018, and are projected to remain so over the medium term. FDI flows are mostly associated with the tourism sector.

- **Financing pressures** are set to increase as amortization repayments on large infrastructure projects are coming due. Debt service indicators are projected to increase, peaking in 2022

because of the US$250 million repayment of a bullet sovereign bond issued in 2017. This represents an important rollover risk considering the possible change in investors' appetite for Maldivian debt under tightening global financial conditions. The Sovereign Development Fund (SDF) was set up to mitigate these rollover risks. In this vein, earmarked funds are built from the Airport Development Fee, revenue from revisions to fees and services provided in the main International Airport, the sovereign guarantee fees, and dividends from the Maldives Airports Company Limited, to repay external loans. Accumulated funds as of end 2018 were US$125 million. Staff projects the SDF to sufficiently cover the international sovereign bond obligations coming due, backed by a government instituted policy of weekly revenue earmarking to the Fund. This supports staff judgment that debt is sustainable.

- **Exports of goods and services** are 60 percent of GDP, making the stress test associated with exports the most severe, along with the combination shock. Market financing, through the issuance of bonds stands at US$350 million to date, around 18 percent of the outstanding PPG external debt. The relatively small share limits risks from external financing shocks so the potential for heightened liquidity needs remains moderate (Figure 3). Gross financing needs surpass their threshold due to the large share of short-term debt in domestic debt issuance.

8. Overall public debt sustainability. Public sector debt stood at an estimated 72 percent of GDP in 2018. Domestic public debt is an important source of financing in the Maldives and is projected to remain so. Despite favorable automatic debt dynamics, the primary deficit will contribute to an increase in public debt over the medium-term reaching 83 percent of GDP in 2021. Staff recommends an ambitious fiscal adjustment that would achieve a primary surplus of 1.4 percent of GDP and would gradually lower public debt to 70 percent of GDP by 2023. The adjustment would reduce fiscal risks, and is designed to minimize the adverse impact on economic growth. Staff also recommends developing the domestic debt market by lengthening debt maturity and minimizing rollover risks from short-term debt.

9. Tailored stress tests for natural disasters, contingent liabilities, and market financing are relevant for the Maldives. The Maldives is susceptible to a host of natural disasters and has issued two sovereign bonds (amounting to US$350 million). The tailored stress tests were kept to their pre-set, default calibrations, and they are not the most extreme shocks for any of the indicators. The most extreme shocks are the combination shock and the shock to exports for the PPG external debt indicators, and the shock to growth for the public debt indicators.

ASSESSMENT

10. The Maldives has a high risk of external debt distress and a high overall risk of debt distress. The debt-to-exports ratio does not breach its threshold under the baseline, but the remaining external debt indicators do. Risks stem from the elevated public debt and the rollover risks from sovereign bonds. The SDF helps mitigate these risks and the projected decline in externally-financed SOE projects puts a cap on these risks, supporting a sustainable granular risk rating. Staff recommends fiscal consolidation to put public debt on a downward path, developing the domestic debt market to allow longer tenors, and

ensuring that the authorities are selective about externally-financed projects and that new external debt is contracted on concessional terms, to the extent possible.

AUTHORITIES' VIEWS

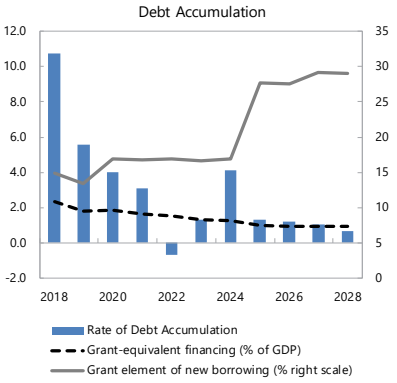
11. The authorities recognize the risk that high public debt poses. The new administration is focusing on close oversight of the issuance of government guarantees and has been publishing data on government guarantees on the Ministry of Finance's website for increased transparency. They maintain that institutional weaknesses are not the primary cause for the increase in government-guaranteed debt owed to China but reflect rather decisions by the previous administration, to finance large infrastructure projects. They emphasized the role of the SDF in mitigating financing risks over the medium term and intend to continue building up funds for repaying debt.

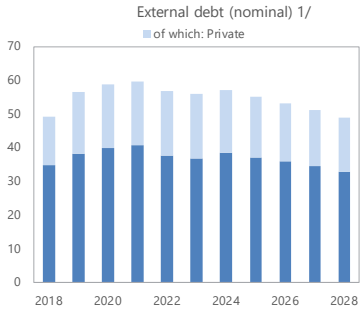
Table 1. Maldives: External Debt Sustainability Framework, Baseline Scenario, 2015-2038

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
External debt (nominal) 1/	**26.7**	**30.3**	**39.5**	**49.3**	**56.6**	**58.7**	**59.7**	**56.9**	**55.9**	**49.0**	**29.6**	**35.3**	**54.8**
of which: public and publicly guaranteed (PPG)	**16.8**	**20.5**	**25.4**	**34.8**	**38.2**	**40.0**	**40.7**	**37.8**	**36.7**	**33.0**	**19.3**	**22.3**	**37.0**
Change in external debt	-2.2	3.7	9.2	9.8	7.3	2.1	1.0	-2.8	-1.0	-2.2	-0.6		
Identified net debt-creating flows	**-2.7**	**11.0**	**8.9**	**12.7**	**7.4**	**2.8**	**0.9**	**-1.0**	**-2.2**	**-3.5**	**-4.2**	**-0.2**	**0.4**
Non-interest current account deficit	**7.3**	**23.1**	**21.7**	**23.0**	**18.0**	**13.0**	**11.2**	**9.3**	**8.0**	**5.4**	**0.6**	**12.3**	**10.5**
Deficit in balance of goods and services	-9.2	1.2	4.6	6.7	0.5	-3.5	-4.7	-6.0	-7.2	-10.2	-15.8	**-6.8**	**-5.4**
Exports	76.7	71.5	67.7	65.3	65.7	65.6	65.7	65.7	65.5	66.0	67.1		
Imports	67.6	72.7	72.3	72.0	66.2	62.1	61.0	59.7	58.3	55.8	51.3		
Net current transfers (negative = inflow)	8.5	14.3	9.2	9.5	9.2	9.0	8.8	8.7	8.5	7.7	7.8	**9.3**	**8.5**
of which: official	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	7.9	7.7	7.9	6.7	8.2	7.5	7.2	6.6	6.7	7.9	8.6	**9.8**	**7.4**
Net FDI (negative = inflow)	**-7.3**	**-10.4**	**-10.1**	**-8.5**	**-9.2**	**-9.2**	**-9.6**	**-9.3**	**-9.1**	**-7.3**	**-3.7**	**-9.5**	**-8.7**
Endogenous debt dynamics 2/	**-2.7**	**-1.7**	**-2.7**	**-1.7**	**-1.4**	**-1.0**	**-0.8**	**-1.0**	**-1.1**	**-1.6**	**-1.1**		
Contribution from nominal interest rate	0.2	0.1	0.2	1.0	1.6	2.1	2.2	2.0	1.8	1.0	0.4		
Contribution from real GDP growth	-0.8	-1.8	-1.9	-2.7	-2.9	-3.2	-3.0	-3.0	-2.9	-2.6	-1.5		
Contribution from price and exchange rate changes	-2.1	0.0	-0.9	...	...	...	...	...	...	...	...		
Residual 3/	**0.4**	**-7.3**	**0.3**	**-3.0**	**-0.1**	**-0.7**	**0.1**	**-1.8**	**1.2**	**1.2**	**3.6**	**-1.1**	**0.5**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**25.4**	**33.0**	**35.5**	**36.3**	**36.5**	**33.1**	**31.9**	**28.4**	**14.5**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**37.5**	**50.4**	**54.0**	**55.4**	**55.5**	**50.4**	**48.6**	**43.0**	**21.6**		
PPG debt service-to-exports ratio	**5.9**	**4.5**	**0.3**	**2.8**	**6.1**	**7.2**	**7.5**	**12.0**	**7.8**	**3.4**	**2.1**		
PPG debt service-to-revenue ratio	**17.3**	**12.0**	**0.7**	**7.1**	**15.8**	**18.5**	**19.6**	**31.2**	**20.1**	**8.6**	**5.1**		
Gross external financing need (Million of U.S. dollars)	350.8	752.1	763.9	1157.4	880.8	700.9	622.8	770.3	520.1	293.7	-50.5		
Key macroeconomic assumptions													
Real GDP growth (in percent)	2.9	7.3	6.9	7.5	6.5	6.0	5.5	5.5	5.5	5.5	5.5	**5.2**	**5.8**
GDP deflator in US dollar terms (change in percent)	7.9	0.1	3.2	2.0	2.0	2.4	2.5	2.5	2.3	2.0	2.0	**4.8**	**2.2**
Effective interest rate (percent) 4/	0.8	0.5	0.7	2.7	3.4	4.0	4.0	3.6	3.4	2.2	1.5	**1.0**	**2.9**
Growth of exports of G&S (US dollar terms, in percent)	-4.7	0.1	4.5	5.8	9.3	8.4	8.4	8.0	7.8	7.8	7.8	**6.7**	**7.9**
Growth of imports of G&S (US dollar terms, in percent)	0.5	15.5	9.9	9.2	-0.2	1.9	6.2	5.8	5.5	6.5	6.7	**9.4**	**5.6**
Grant element of new public sector borrowing (in percent)	...	...	...	14.9	13.4	16.9	16.8	16.9	16.7	29.1	29.0	**...**	**20.5**
Government revenues (excluding grants, in percent of GDP)	26.1	27.0	26.8	26.0	25.4	25.4	25.3	25.3	25.4	26.1	27.9	**23.5**	**25.6**
Aid flows (in Million of US dollars) 5/	41.3	12.2	14.1	103.6	84.3	150.1	149.2	154.7	141.0	202.4	411.8		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	2.3	1.8	1.8	1.6	1.5	1.3	0.9	0.8	**...**	**1.4**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	17.9	19.4	23.1	23.2	23.4	23.7	38.7	33.6	**...**	**28.3**
Nominal GDP (Million of US dollars)	4,098	4,402	4,858	5,328	5,786	6,283	6,795	7,348	7,937	11,451	23,896		
Nominal dollar GDP growth	11.0	7.4	10.4	9.7	8.6	8.6	8.1	8.1	8.0	7.6	7.6	**10.2**	**8.1**
Memorandum items:													
PV of external debt 7/	...	...	39.5	47.4	53.9	55.1	55.5	52.2	51.0	44.4	24.9		
In percent of exports	...	...	58.4	72.6	81.9	84.0	84.4	79.6	77.9	67.2	37.0		
Total external debt service-to-exports ratio	11.0	6.0	6.1	11.1	9.8	11.2	11.4	15.9	11.7	6.7	4.2		
PV of PPG external debt (in Million of US dollars)			1233.9	1755.6	2052.4	2283.6	2477.6	2432.2	2529.0	3253.2	3467.8		
(PVt-PVt-1)/GDPt-1 (in percent)				10.7	5.6	4.0	3.1	-0.7	1.3	0.7	0.9		
Non-interest current account deficit that stabilizes debt ratio	9.5	19.4	12.5	13.2	10.6	10.9	10.3	12.1	9.0	7.6	1.2		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

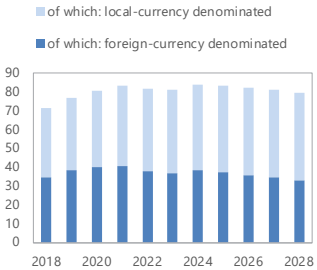
Table 2. Maldives: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015-2038

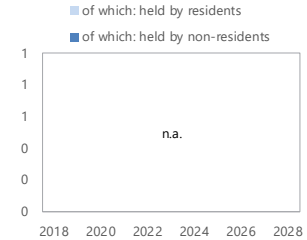
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
Public sector debt 1/	**53.4**	**59.4**	**61.6**	**71.4**	**76.8**	**80.7**	**83.1**	**81.5**	**81.4**	**79.6**	**53.8**	**53.7**	**80.5**
of which: external debt	**16.8**	**20.5**	**25.4**	**34.8**	**38.2**	**40.0**	**40.7**	**37.8**	**36.7**	**33.0**	**19.3**	**22.3**	**37.0**
Change in public sector debt	-1.8	6.0	2.2	9.8	5.4	3.9	2.4	-1.6	-0.1	-1.7	-2.0		
Identified debt-creating flows	**0.6**	**5.6**	**-2.6**	**-0.9**	**0.7**	**1.3**	**0.9**	**0.3**	**-0.1**	**-1.5**	**-2.9**	**2.4**	**-0.3**
Primary deficit	**3.9**	**7.4**	**1.5**	**2.6**	**3.0**	**3.2**	**2.6**	**2.2**	**1.8**	**0.6**	**-1.4**	**6.1**	**1.9**
Revenue and grants	27.1	27.3	27.1	26.5	26.0	26.0	25.8	25.8	25.8	26.4	28.1	**24.3**	**26.1**
of which: grants	1.0	0.3	0.3	0.5	0.6	0.6	0.5	0.5	0.5	0.3	0.2		
Primary (noninterest) expenditure	31.1	34.7	28.6	29.1	29.0	29.2	28.4	28.0	27.6	27.1	26.7	**30.5**	**28.0**
Automatic debt dynamics	**-3.4**	**-1.8**	**-4.1**	**-3.5**	**-2.2**	**-1.9**	**-1.7**	**-2.0**	**-1.8**	**-2.1**	**-1.5**		
Contribution from interest rate/growth differential	-2.3	-1.9	-3.8	-3.5	-0.6	-1.8	-1.6	-1.9	-1.8	-2.2	-1.6		
of which: contribution from average real interest rate	-0.7	1.7	0.0	0.8	3.8	2.6	2.6	2.4	2.4	2.0	1.4		
of which: contribution from real GDP growth	-1.5	-3.6	-3.8	-4.3	-4.3	-4.4	-4.2	-4.3	-4.3	-4.2	-2.9		
Contribution from real exchange rate depreciation	-1.1	0.2	-0.3	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.7**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-2.4**	**0.4**	**4.8**	**10.7**	**3.0**	**2.5**	**1.5**	**-1.9**	**-0.1**	**-0.1**	**0.9**	**0.2**	**1.8**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**61.6**	**69.5**	**74.0**	**77.1**	**78.9**	**76.8**	**76.5**	**75.0**	**49.1**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**227.7**	**262.4**	**284.1**	**296.8**	**305.3**	**297.9**	**296.3**	**283.7**	**174.7**		
Debt service-to-revenue and grants ratio 3/	**72.6**	**69.3**	**70.5**	**76.7**	**70.8**	**64.9**	**61.1**	**82.8**	**86.4**	**69.8**	**47.1**		
Gross financing need 4/	21.6	24.5	20.7	21.1	17.4	15.3	13.4	15.7	19.0	16.9	10.5		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	2.9	7.3	6.9	7.5	6.5	6.0	5.5	5.5	5.5	5.5	5.5	**5.2**	**5.8**
Average nominal interest rate on external debt (in percent)	1.3	0.8	1.0	4.2	4.8	6.0	5.8	5.3	5.2	3.2	2.3	**1.6**	**4.4**
Average real interest rate on domestic debt (in percent)	-1.9	5.1	0.4	0.9	3.5	3.2	3.2	3.2	3.5	3.9	3.9	**-0.3**	**3.3**
Real exchange rate depreciation (in percent, + indicates depreciation)	-6.0	1.0	-1.6	...	...	...	...	...	...	...	...	**-2.8**	**...**
Inflation rate (GDP deflator, in percent)	7.9	0.1	3.2	2.0	2.0	2.4	2.5	2.5	2.3	2.0	2.0	**6.7**	**2.2**
Growth of real primary spending (deflated by GDP deflator, in percent)	3.0	19.9	-12.0	9.3	6.2	6.6	2.9	3.9	3.9	5.2	5.5	**4.2**	**5.3**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	5.7	1.4	-0.7	-7.2	-2.4	-0.7	0.2	3.8	1.9	2.3	0.7	**2.1**	**0.3**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security, central bank, government-guaranteed debt. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

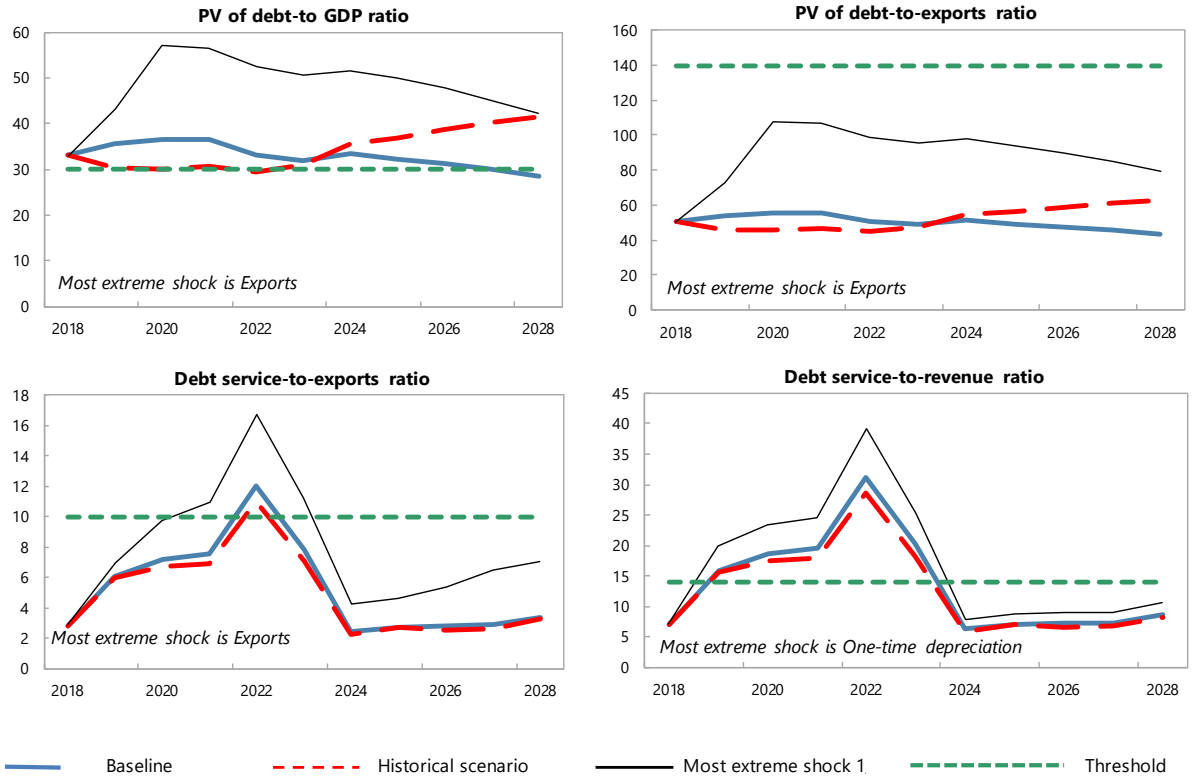
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Maldives Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2018-2028



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	No	No
Commodity Prices 2/	n.a.	n.a.
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	3.0%	3.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	6	6

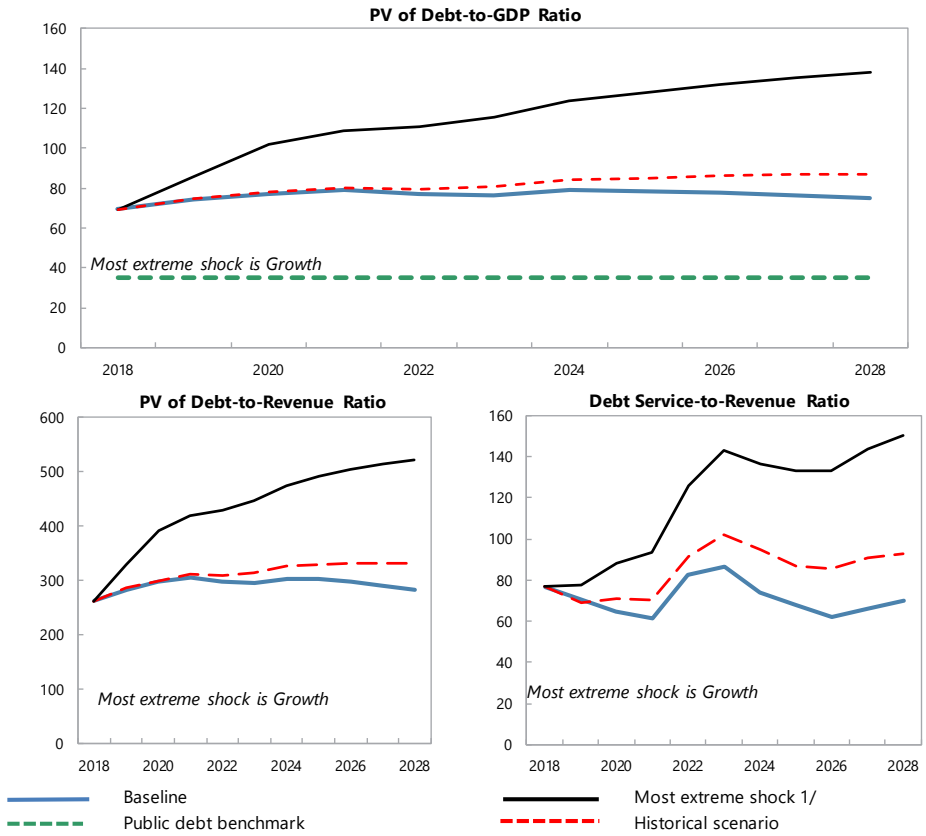
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Maldives: Indicators of Public Debt Under Alternatives Scenarios, 2018-2028



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	23%	23%
Domestic medium and long-term	43%	43%
Domestic short-term	35%	35%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	3.0%	3.0%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	3.8%	3.8%
Avg. maturity (incl. grace period)	5	5
Avg. grace period	3	3
Domestic short-term debt		
Avg. real interest rate	3%	3.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Maldives: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018-2028

(In percent)

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	**33**	**35**	**36**	**36**	**33**	**32**	**33**	**32**	**31**	30	28
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	**33**	**30**	**30**	**31**	29	**31**	**36**	**37**	**39**	**40**	**42**
B. Bound Tests											
B1. Real GDP growth	**33**	**39**	**44**	**44**	**40**	**38**	**40**	**39**	**37**	**36**	**34**
B2. Primary balance	**33**	**40**	**46**	**48**	**46**	**47**	**50**	**50**	**49**	**49**	**48**
B3. Exports	**33**	**43**	**57**	**56**	**52**	**50**	**52**	**50**	**48**	**45**	**42**
B4. Other flows 3/	**33**	**39**	**44**	**44**	**40**	**39**	**40**	**39**	**37**	**35**	**33**
B5. One-time 30 percent nominal depreciation	**33**	**45**	**45**	**45**	**41**	**39**	**41**	**40**	**38**	**37**	**35**
B6. Combination of B1-B5	**33**	**46**	**50**	**51**	**47**	**45**	**47**	**46**	**44**	**42**	**40**
C. Tailored Tests											
C1. Combined contingent liabilities	**33**	**37**	**40**	**41**	**40**	**40**	**43**	**43**	**43**	**43**	**43**
C2. Natural disaster	**33**	**39**	**41**	**43**	**41**	**42**	**45**	**46**	**46**	**46**	**46**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**33**	**40**	**42**	**42**	**39**	**37**	**38**	**36**	**34**	**33**	**31**
Threshold	30	30	30	30	30	30	30	30	30	30	30
PV of debt-to-exports ratio											
Baseline	50	54	55	55	50	49	51	49	47	45	43
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	50	46	46	47	45	47	54	56	58	61	63
B. Bound Tests											
B1. Real GDP growth	50	54	55	55	50	49	51	49	47	45	43
B2. Primary balance	50	61	70	73	70	71	75	75	75	74	73
B3. Exports	50	73	108	107	99	96	98	94	90	85	79
B4. Other flows 3/	50	60	67	67	61	59	61	59	56	54	50
B5. One-time 30 percent nominal depreciation	50	54	54	54	49	48	50	48	46	45	42
B6. Combination of B1-B5	50	68	66	77	71	69	71	69	66	63	60
C. Tailored Tests											
C1. Combined contingent liabilities	50	57	61	63	61	62	66	66	66	65	65
C2. Natural disaster	50	60	64	66	64	66	71	71	71	71	70
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	50	55	57	58	53	51	52	49	47	45	42
Threshold	140	140	140	140	140	140	140	140	140	140	140
Debt service-to-exports ratio											
Baseline	3	6	7	8	**12**	8	2	3	3	3	3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	3	6	7	7	**11**	7	2	3	3	3	3
B. Bound Tests											
B1. Real GDP growth	3	6	7	8	**12**	8	2	3	3	3	3
B2. Primary balance	3	6	7	8	**13**	9	3	4	4	4	5
B3. Exports	3	7	10	**11**	**17**	**11**	4	5	5	7	7
B4. Other flows 3/	3	6	7	8	**12**	8	3	3	3	4	4
B5. One-time 30 percent nominal depreciation	3	6	7	8	**12**	8	2	3	3	3	3
B6. Combination of B1-B5	3	7	9	10	**15**	10	4	4	5	5	5
C. Tailored Tests											
C1. Combined contingent liabilities	3	6	7	8	**12**	8	3	3	3	3	4
C2. Natural disaster	3	6	8	8	**13**	8	3	3	4	4	4
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	3	6	7	8	**13**	**11**	8	7	5	3	3
Threshold	10	10	10	10	10	10	10	10	10	10	10
Debt service-to-revenue ratio											
Baseline	7	**16**	**19**	**20**	**31**	**20**	6	7	7	7	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	7	**16**	**17**	**18**	**29**	**18**	6	7	7	7	8
B. Bound Tests											
B1. Real GDP growth	7	**17**	**22**	**23**	**37**	**24**	7	8	9	9	10
B2. Primary balance	7	**16**	**19**	**21**	**33**	**22**	8	9	10	11	13
B3. Exports	7	**16**	**20**	**23**	**35**	**23**	9	10	11	13	**14**
B4. Other flows 3/	7	**16**	**19**	**21**	**32**	**21**	7	8	9	10	11
B5. One-time 30 percent nominal depreciation	7	**20**	**23**	**25**	**39**	**25**	8	9	9	9	11
B6. Combination of B1-B5	7	**17**	**23**	**25**	**39**	**26**	9	10	12	12	13
C. Tailored Tests											
C1. Combined contingent liabilities	7	**16**	**19**	**20**	**32**	**21**	7	8	9	9	10
C2. Natural disaster	7	**16**	**19**	**20**	**32**	**21**	8	9	9	9	11
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	7	**16**	**19**	**21**	**33**	**29**	**20**	**18**	13	7	8
Threshold	14	14	14	14	14	14	14	14	14	14	14

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

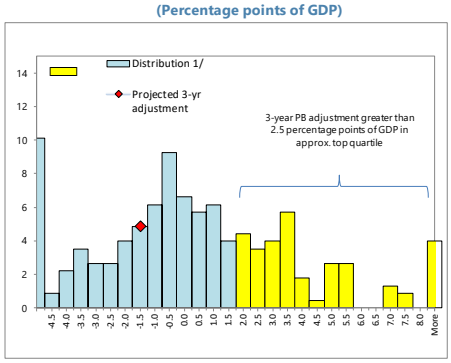
Table 4. Maldives: Sensitivity Analysis for Key Indicators of Public Debt, 2018-2028

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	**70**	**74**	**77**	**79**	**77**	**77**	**79**	**78**	**78**	**77**	**75**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	**70**	**75**	**78**	**80**	**80**	**81**	**84**	**85**	**86**	**87**	**87**
B. Bound Tests											
B1. Real GDP growth	**70**	**86**	**102**	**109**	**111**	**116**	**123**	**128**	**132**	**136**	**138**
B2. Primary balance	**70**	**83**	**96**	**97**	**95**	**95**	**98**	**98**	**97**	**97**	**95**
B3. Exports	**70**	**81**	**96**	**97**	**94**	**93**	**95**	**94**	**92**	**90**	**87**
B4. Other flows 3/	**70**	**78**	**85**	**86**	**84**	**83**	**85**	**85**	**84**	**82**	**80**
B5. One-time 30 percent nominal depreciation	**70**	**81**	**81**	**81**	**77**	**75**	**76**	**75**	**73**	**72**	**69**
B6. Combination of B1-B5	**70**	**80**	**90**	**93**	**91**	**92**	**96**	**97**	**97**	**97**	**97**
C. Tailored Tests											
C1. Combined contingent liabilities	**70**	**80**	**83**	**85**	**83**	**83**	**86**	**86**	**86**	**86**	**85**
C2. Natural disaster	**70**	**85**	**88**	**90**	**88**	**88**	**92**	**92**	**92**	**92**	**91**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**70**	**74**	**78**	**80**	**78**	**78**	**80**	**78**	**77**	**76**	**74**
Public debt benchmark	35	35	35	35	35	35	35	35	35	35	35
PV of Debt-to-Revenue Ratio											
Baseline	262	284	297	305	298	296	304	302	297	291	284
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	262	287	300	311	309	314	326	329	331	332	331
B. Bound Tests											
B1. Real GDP growth	262	329	391	419	428	446	474	491	504	514	522
B2. Primary balance	262	320	368	376	369	368	377	376	373	367	360
B3. Exports	262	310	368	374	364	360	366	362	354	342	329
B4. Other flows 3/	262	299	326	333	325	323	329	326	320	312	302
B5. One-time 30 percent nominal depreciation	262	313	314	314	299	292	294	289	282	272	262
B6. Combination of B1-B5	262	306	347	359	354	357	370	373	372	370	366
C. Tailored Tests											
C1. Combined contingent liabilities	262	308	320	328	321	322	332	332	330	326	321
C2. Natural disaster	262	324	337	347	341	342	353	354	353	350	345
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	262	286	300	311	304	302	307	302	296	289	281
Debt Service-to-Revenue Ratio											
Baseline	77	71	65	61	83	86	74	68	62	66	70
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	77	69	71	70	92	102	95	87	85	91	93
B. Bound Tests											
B1. Real GDP growth	77	78	88	93	126	143	137	133	133	144	151
B2. Primary balance	77	71	83	88	106	120	116	108	101	105	109
B3. Exports	77	71	66	64	85	89	76	70	66	71	75
B4. Other flows 3/	77	71	65	62	84	87	75	69	64	68	72
B5. One-time 30 percent nominal depreciation	77	69	71	72	100	108	92	84	81	88	90
B6. Combination of B1-B5	77	70	73	82	105	116	108	103	100	105	108
C. Tailored Tests											
C1. Combined contingent liabilities	77	71	81	75	99	114	107	96	91	97	101
C2. Natural disaster	77	72	87	79	102	120	113	102	96	103	107
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	77	71	66	62	85	95	87	79	67	66	70

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

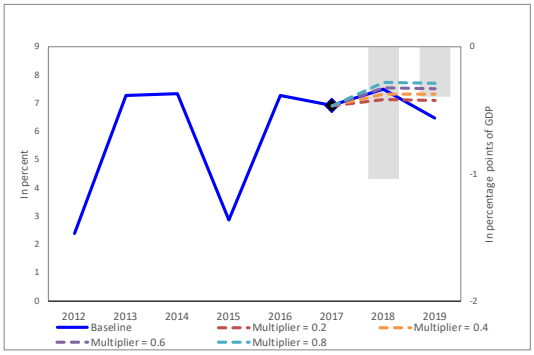
Figure 3. Maldives: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



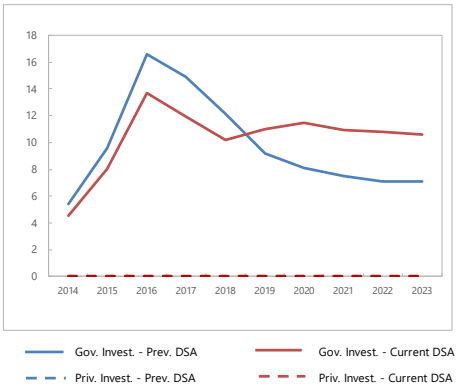
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

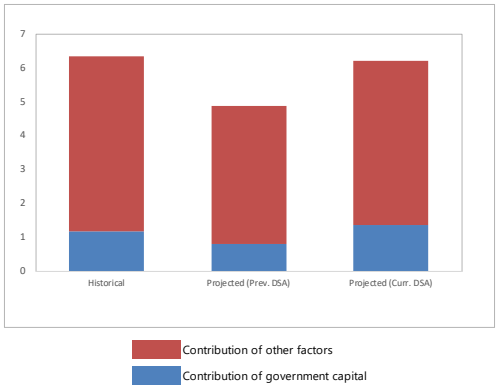
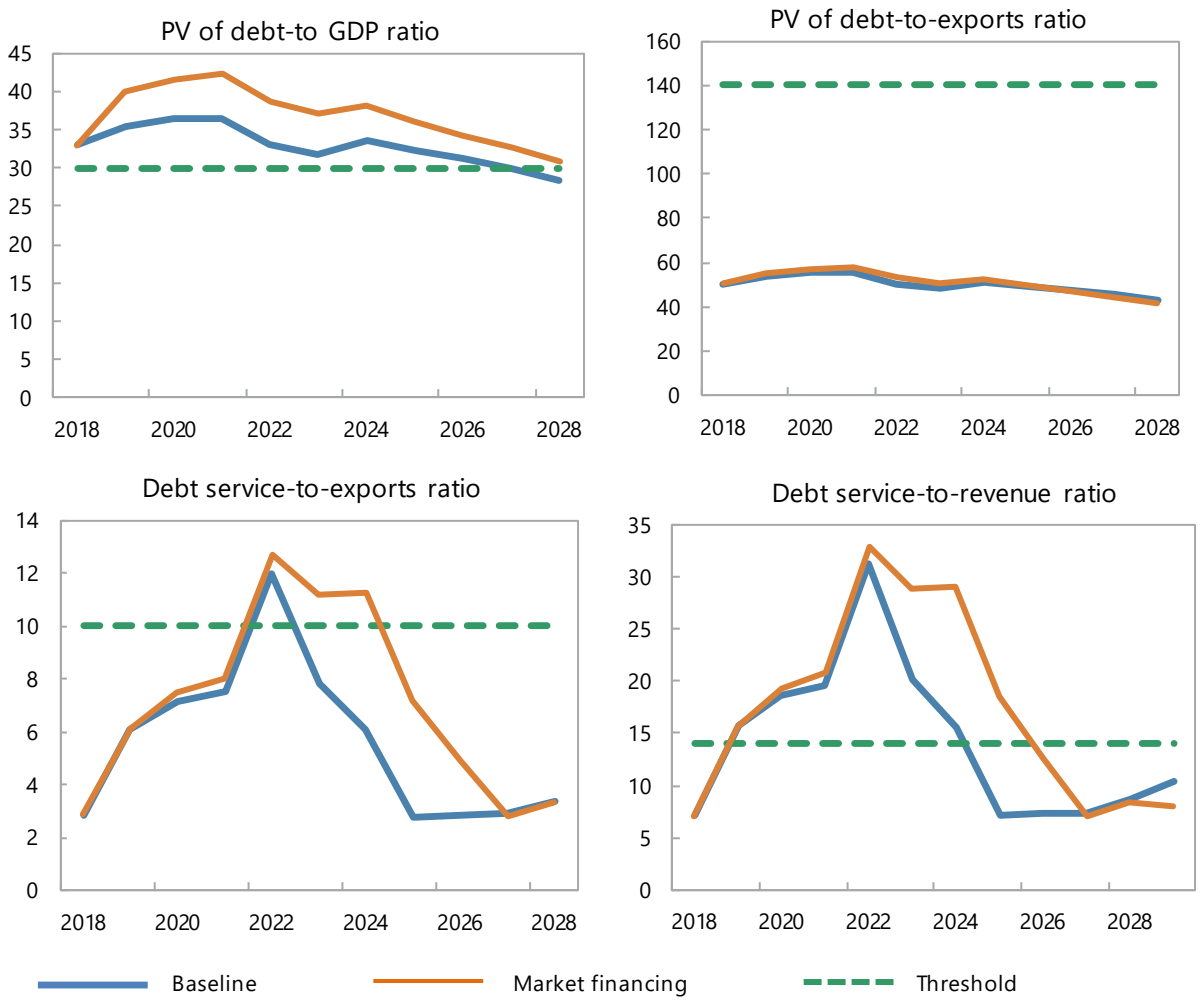


Figure 4. Maldives: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	21	350
Breach of benchmark	Yes	No
Potential heightened liquidity needs	Moderate	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.